Generative Modelling for Realistic Mars Landing Site Visualisation: A Critical Survey of the Enabling Literature

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***Abstract*—Preparing rover operators for planetary surface missions demands photorealistic virtual training environments that replicate the target landing site at the resolution rover cameras actually deliver. For Mars, the highest-resolution orbital resource is the High Resolution Imaging Science Experiment (HiRISE) at 25 cm/pixel in nadir view, whereas rover cameras resolve sub-millimetre surface texture from near-grazing perspective angles, a gap of roughly three orders of magnitude that standard single-image super-resolution methods, which upscale at most $4\times$, cannot close. This review critically examines twenty publications spanning single-image super-resolution, image-to-image domain translation, digital terrain model generation, neural radiance field rendering, and virtual reality reconstruction, evaluating each against the specific challenge of synthesising rover-perspective surface textures from nadir orbital imagery for mission training simulation. The analysis shows that while each sub-field has advanced substantially since 2021 transformer-based and diffusion-based super-resolution outperform earlier GAN methods; NeRF representations of Mars surface have been demonstrated from rover imagery; and physics-constrained generative networks reduce terrain reconstruction error by more than 50% no published method jointly addresses the view-angle transformation and high-fidelity texture synthesis within a single framework. The three primary barriers are the absence of aligned cross-view training data, the documented risk of hallucination in diffusion-model outputs for mission-critical visualisations, and the computational cost of diffusion-based synthesis at orbital-to-rover scale. These findings delineate the specific research gap that motivates the proposed dissertation.**

***Index Terms*—Mars surface imagery, super-resolution, image-to-image translation, neural radiance fields, digital terrain models, diffusion models, virtual reality simulation, HiRISE, domain adaptation, planetary exploration**

I. Introduction

Successful deployment of a planetary rover begins not with launch, but with exhaustive ground simulation. The ESA ExoMars Rosalind Franklin Mission (RFM), targeting Oxia Planum as its landing site, requires training environments sufficiently faithful to the actual terrain to rehearse traverse planning, instrument operations, and hazard response. Virtual reality (VR) satisfies this requirement in principle, but its operational value collapses when surface textures become visually implausible. A crew navigating a smoothed, geometrically approximate mesh is not rehearsing the same perceptual and decision-making conditions it will face on the ground. The environment must look like Mars at the scale rover cameras can actually resolve.

The core difficulty is a data paradox embedded in planetary imaging architecture. The highest-resolution orbital survey instrument on Mars is the High Resolution Imaging Science Experiment (HiRISE) aboard NASA's Mars Reconnaissance Orbiter, which acquires imagery at 25–50 cm/pixel in a nadir (directly downward) viewing geometry [1]. The ExoMars Trace Gas Orbiter's Colour and Stereo Surface Imaging System (CaSSIS) provides broader multi-spectral coverage at a coarser 4–5 m/pixel [2]. The Mastcam-Z (Rover camera) aboard NASA's Perseverance rover being the most capable current example resolve individual regolith grains at sub-millimetre scale from a vantage point roughly 2 m above the surface, looking across it at oblique angles. The gap between HiRISE's 25 cm/pixel and sub-millimetre rover resolution spans approximately three orders of magnitude in linear scale, or six orders of magnitude in area. Current single-image super-resolution (SISR) methods reduce this gap by at most a factor of four [2], leaving it essentially intact.

Resolution alone, however, does not capture the full dimensionality of the problem. A nadir HiRISE image of a boulder field depicts the horizontal top surfaces of rocks. A rover standing 10 m away sees the same boulders in perspective: vertical faces, shadows cast across the regolith floor, and the parallax between foreground grains and background outcrops. Upsampling a nadir image, however aggressively, cannot hallucinate these perspective-view features because they are geometrically absent from the input. The view-angle transformation from nadir to perspective requires inferring three-dimensional surface geometry from two-dimensional orbital reflectance data and re-rendering that geometry from a different camera position is a problem that SR, by construction, does not address.

This review surveys twenty publications across seven thematic areas, assessing each against the specific research requirement: generating rover-perspective surface texture from nadir orbital imagery at VR-grade fidelity for pre-mission simulation. The seven thematic areas are:

1) Orbital imaging systems and Mars data characterisation;
2) Single-image super-resolution for planetary remote sensing;
3) Image-to-image translation frameworks applicable to view-domain bridging;

4) Monocular digital terrain model (DTM) generation from orbital imagery;
5) Neural radiance field (NeRF) and Gaussian splatting methods applied to Mars;
6) Virtual reality reconstruction pipelines for planetary mission simulation;
7) Hallucination and epistemic reliability in generative models used for scientific inference.

The critical finding, developed throughout the analysis, is that no published method and no obvious composition of published methods, closes the nadir-orbital to rover-perspective gap within a single trainable framework. The dissertation work targets precisely this gap.

II. Taxonomy and Conceptual Framework

The literature reviewed here does not constitute a unified research programme. It comprises several independently advancing communities like remote sensing image processing, computer vision generative modelling, planetary science, and VR engineering whose outputs are jointly necessary to the downstream VR synthesis requirement without any of them addressing it explicitly. Three axes organise the territory.

The first axis is *spatial view*: methods operate in the nadir domain (orbital, top-down), the perspective domain (rover, oblique), or across both. The second axis is *task type*: methods either enhance existing imagery (super-resolution, inpainting), translate between domains (image-to-image translation, domain adaptation), or recover and render geometric structure (DTM generation, NeRF, Gaussian splatting). The third axis is *output modality*: visual RGB texture, elevation/depth maps, or rendered video sequences.

Seven clusters emerge from this structure. **Cluster 1** (Orbital data characterisation) establishes baseline understanding of Mars instrument capabilities and their information limits [3], [4]. **Cluster 2** (Nadir SR) comprises methods that upscale orbital images while remaining in the nadir domain [2], [5]–[8]. **Cluster 3** (Image-to-image translation) contains domain-agnostic frameworks that map between arbitrary distributions and could, in principle, translate nadir to perspective [9], [10]. **Cluster 4** (Monocular DTM) covers methods that infer 3D topography from single orbital images [1], [11], [12]. **Cluster 5** (Neural rendering) encompasses NeRF and Gaussian splatting methods applied to Mars scenes [13]–[17]. **Cluster 6** (VR reconstruction) covers end-to-end pipelines targeting mission simulation environments [14], [18]. **Cluster 7** (Reliability) analyses hallucination and epistemic validity in generative scientific models [19], [20].

The gap between Cluster 2 and Cluster 5 is the primary concern of this dissertation. Nadir SR methods keep the viewpoint fixed; neural rendering methods require perspective imagery as input. No reviewed method translates from nadir to perspective in a way that synthesises rover-viewable texture from orbital data. This structural gap is not an oversight in the literature; it reflects the distinct communities that have developed these tools and the absence of a formal problem statement that spans both.

III. Critical Analysis of the Literature

A. The Orbital Data Landscape and Its Inherent Constraints

HiRISE provides the sharpest available view of the Martian surface from orbit, yet its coverage is drastically limited. Fewer than 3% of the Martian surface has been imaged at 25 cm/pixel resolution, because the instrument's 6 km swath width and the competing demands of the science team constrain acquisition rates [1]. CaSSIS covers broader areas at 4–5 m/pixel, and the Context Camera (CTX) on MRO provides near-global coverage at 6 m/pixel. Each instrument occupies a distinct rung in the resolution hierarchy, and the gulf between even the finest orbital product and rover cameras remains unbridged regardless of which instrument is the starting point.

Serrano *et al.* [3] expose a revealing consequence of this hierarchy. The paper trains a Bayesian network to infer HiRISE-equivalent rock density, a key landing hazard metric from coarser CTX image features, incorporating geomorphic unit labels as prior information. The model succeeds in probabilistic density estimation but does so without reconstructing surface *appearance*: it predicts rock abundance, not rock texture or shape. This is not a design limitation; it is a statement about information content. CTX pixels do not contain the sub-pixel spectral and spatial variation from which HiRISE-equivalent visual texture could be deterministically reconstructed. Any generative model that attempts to synthesise rover-scale visual texture from nadir orbital imagery faces the same information-theoretic constraint. The synthesis is necessarily generative and grounded in statistical priors rather than purely reconstructive.

Osadnik *et al.* [4] respond to data scarcity by releasing MCTED, comprising 80,898 paired CTX orthoimage–digital elevation model (DEM) samples derived from the Day *et al.* processing pipeline, with tooling to handle common artefacts and missing-data regions. The contribution is genuine: standardised, ML-ready paired data has been conspicuously absent from Mars terrain modelling research, and MCTED provides a reproducible training benchmark. Its limitation, from the perspective of this dissertation, is structural: the pairs link nadir images to elevation maps, not nadir images to perspective views. A dataset providing the same surface patch in both nadir orbital and rover perspective at aligned resolution does not yet exist, and its absence constrains every approach in Clusters 3 and 5 as much as it constrains novel methods.

B. Super-Resolution for Planetary Remote Sensing

SR research on planetary imagery has progressed through three generations since 2021: GAN-based methods modelled closely on terrestrial benchmarks, attention and transformer architectures that better capture long-range spatial dependencies, and diffusion-based synthesis that trades pixel-level reconstruction accuracy for perceptual realism.

1) GAN-Based SR: Tao *et al.* [2] propose MARSGAN, an enhancement of ESRGAN incorporating adaptive-weighted residual dense blocks (AW-RRDB) with Gaussian noise injection and a multi-scale reconstruction scheme that uses pixel-shuffling to suppress checkerboard artefacts. Trained exclusively on HiRISE images and applied at inference to

CaSSIS tiles, MARSGAN achieves approximately 3$\times$ effective resolution improvement, reducing apparent CaSSIS resolution from 4–5 m/pixel to roughly 1.3 m/pixel. The work is evaluated across eight geologically diverse Mars science sites including the Mars 2020 Perseverance landing site at Jezero Crater. Notably, however, the evaluation is entirely qualitative, no PSNR or SSIM scores are reported. This is not merely a presentation choice. Co-registering CaSSIS and HiRISE acquisitions of the same location precisely enough for pixel-level comparison is non-trivial given the instruments' different viewing geometries, acquisition times, and spatial sampling. The qualitative approach is honest but leaves cross-method performance comparison opaque, a problem that recurs throughout this sub-field.

2) Transformer-Based SR: Wu *et al.* [5] address SR at $\times 2$, $\times 4$, and $\times 8$ magnification through RSTSRN, which extends LapSRN's Laplacian pyramid architecture with recursive Swin Transformer attention blocks. The recursive strategy processes a single input through shared attention weights at multiple scales, reducing parameter count while expanding the effective receptive field. On Mars image benchmarks, RSTSRN improves upon LapSRN by 0.35 dB PSNR at $\times 2$, 0.88 dB at $\times 4$, and 1.22 dB at $\times 8$, with corresponding SSIM gains of 0.0048, 0.0114, and 0.0149, and outperforms competing transformer architectures including SwinIR. The improvement at higher scale factors is the most operationally relevant finding: the larger the SR factor, the more the texture generation task resembles free synthesis rather than reconstruction, and transformer self-attention mechanisms are better equipped than convolutional filters to maintain long-range consistency across synthesised regions. The critical flaw is that evaluation uses bicubic downsampling to generate low-resolution inputs from the original images, then measures reconstruction against those originals. This protocol assesses recovery from one specific degradation model. Real CaSSIS or CTX pixels do not approximate bicubically downsampled HiRISE, they emerge from a physically distinct optical system with its own point spread function, noise floor, and spectral response. The trained model has never encountered the actual low-to-high resolution relationship it is meant to solve.

3) Diffusion-Based SR and Generative Foundation Models: Khanna *et al.* [7] step back from Mars-specific engineering to propose DiffusionSat, a generative foundation model for satellite imagery broadly. Trained on a heterogeneous collection of public remote sensing datasets including NAIP (1 m/pixel aerial imagery of the USA), xView, FMOW, and SpaceNet, DiffusionSat conditions generation on geolocation metadata and sensor parameters through a 3D positional encoding scheme applied to the diffusion model's conditioning stack. The model achieves competitive FID scores on multi-sensor satellite image generation tasks and demonstrates, crucially, that a single diffusion model can generalise across diverse Earth observation sensors without sensor-specific fine-tuning. The indirect relevance to Mars is significant: DiffusionSat shows that multi-sensor domain generalisation is tractable within a diffusion framework. Its failure mode for Mars is equally clear: its training corpus contains no Mars data; its geolocation conditioning assumes Earth coordinates; and its multi-spectral handling assumes terrestrial atmospheric spectral profiles. Transferring DiffusionSat to Mars would require either large-scale domain fine-tuning or an architectural reformulation of the metadata conditioning that replaces Earth-specific priors with Martian photometric models.

Zhu *et al.* [8] address a tension that DiffusionSat sidesteps: unconditional diffusion models generate perceptually sharp outputs but diverge from pixel-level ground truth, while deterministic regression models achieve high PSNR at the cost of blurred high-frequency texture. Their approach introduces a frequency-decomposition conditioning mechanism that anchors the diffusion denoising trajectory to low-frequency structural content from the input, allowing high-frequency texture to be generated stochastically without compromising large-scale geometry. Evaluated on the DOTA and DIOR-R aerial imagery benchmarks at $\times 4$ SR, the method improves PSNR over prior diffusion SR methods while maintaining competitive FID scores. For Mars applications, the architectural insight translates directly: any synthesis pipeline that must preserve large-scale geomorphology (crater rims, dune crests, valley networks) while generating sub-metre surface texture will encounter exactly this fidelity-perceptual realism trade-off, and frequency decomposition offers a principled way to navigate it.

Lu and Su [6] take a multi-source fusion approach to SR on Mars thermal infrared imagery from the THEMIS instrument. By fusing visible-channel orbital images and MOLA topographic data to guide thermal channel upsampling, the method leverages the complementary spatial information carried by each modality. The approach demonstrates measurable PSNR gains on THEMIS data. Its relevance to the dissertation is indirect: thermal imagery is not the visual RGB texture domain required for photorealistic VR, and the multi-source fusion strategy while promising requires co-registered multi-modal data at the inference site, which may not be uniformly available across candidate Mars landing areas.

Collectively, the SR literature demonstrates that $\times 4$ resolution enhancement of orbital Mars imagery is achievable with GAN, transformer, or diffusion backbones, and that diffusion-based methods best handle the texture stochasticity of geological surfaces. However, $\times 4$ closes a factor-of-four fraction of a three-order-of-magnitude gap. More fundamentally, every reviewed SR method operates nadir-to-nadir: the input is an orbital image; the output is a higher-resolution orbital image. The perspective transformation that converts nadir imagery into what a rover would see is nowhere addressed.

C. Image-to-Image Translation as a Domain-Bridging Mechanism

Image-to-image translation methods map entire image domains to each other, a capability structurally better suited to the nadir-to-perspective view transformation than SR, which merely sharpens detail within a fixed viewpoint. Two principled frameworks appear in the reviewed literature, both grounded in stochastic differential equation (SDE) dynamics.

Kim *et al.* [9] propose UNSB (Unpaired Neural Schrödinger Bridge), which reframes unpaired domain translation as learning an SDE that transports samples from one arbitrary distribution to another. Unlike standard diffusion models, which route data through a Gaussian noise prior and back, the Schrödinger Bridge transports directly between two target distributions. This eliminates the Gaussian prior assumption that forces standard diffusion I2I methods to destroy domain-specific structure before rebuilding it. UNSB addresses the scalability of the Schrödinger Bridge formulation by decomposing the SDE into a sequence of adversarial sub-problems, allowing high-resolution unpaired I2I to succeed where prior SB formulations stalled. On standard benchmarks including horse-to-zebra, cat-to-dog, summer-to-winter, and day-to-night translations, UNSB achieves lower FID than CycleGAN, CUT, and related methods. The architecture is compelling for the Mars nadir-to-perspective problem precisely because it operates on unpaired data: a source domain of HiRISE nadir patches and a target domain of rover perspective images are both available, and they do not need to be geographically co-registered. However, UNSB has been evaluated only on natural image colour-domain translations. The photometric differences between a nadir view and a perspective view of the same Mars surface far exceed the hue and seasonal saturation shifts of the tested benchmarks; they involve changes in shadow geometry, rock face visibility, foreshortening, and inter-object occlusion. All of which require implicit 3D reasoning that the SDE formulation does not explicitly encode.

Xiao *et al.* [10] address the complementary problem: stochastic I2I methods produce different outputs from the same input on successive runs, which is unacceptable in scientific visualisation. Their Dual-approx Bridge exploits Brownian bridge dynamics with two separate neural approximators, one for the forward noising process, one for the reverse to achieve deterministic I2I: the same input produces the same output every time. On paired image generation and super-resolution benchmarks, Dual-approx Bridge achieves superior SSIM and LPIPS compared to stochastic diffusion I2I baselines while maintaining image fidelity comparable to the ground truth. Determinism matters greatly in the Mars context: if two planetary scientists run the same generation pipeline and obtain different reconstructions of the same surface patch, neither output can be used as mission-planning ground truth. Dual-approx Bridge provides the architectural template for a reproducible synthesis system, though, like UNSB, it has not been applied to remote sensing or view-angle translation tasks.

The gap here is categorical rather than incremental. No I2I method has been applied to the nadir-to-perspective Mars translation problem. Neither framework has been tested on remote sensing imagery. The photometric complexity of the view-angle transformation is qualitatively different from the colour and seasonal domain shifts these methods were designed and benchmarked on. Whether SB-based I2I is expressive enough to capture geometry-dependent viewpoint changes or whether a geometry-aware architecture is necessary remains an open empirical question.

D. Digital Terrain Model Generation from Monocular Orbital Imagery

DTM generation from orbital imagery occupies a structurally intermediate position: it extracts 3D geometric information from nadir images, which is a necessary precursor to perspective-view synthesis but does not itself complete the journey to VR-ready visual texture.

Tao *et al.* [1] present MADNet 2.0, which retrieves pixel-scale topography from a single HiRISE image by exploiting photoclinometric cues, the relationship between pixel intensity, surface slope, and sun angle at acquisition. A deep learning model trained on paired HiRISE image–MOLA altimetry data infers a dense elevation map at HiRISE's native 25 cm/pixel resolution. This monocularity is the key engineering advantage: stereo-based DTM generation requires two overlapping acquisitions of the same location, which covers less than 1% of the HiRISE archive; MADNet 2.0 operates on any single-strip observation. The limitation for VR is categorical: the DTM provides surface geometry, not surface appearance. Rocks with identical elevation profiles but different mineralogy, grain size, or weathering state produce entirely different visual textures, none of which MADNet 2.0 recovers.

Zou *et al.* [11] address the photometric ambiguity that confounds DTM generation through PFMGAN, which integrates an explicit solar geometry model and an albedo-aware attention module into a GAN architecture. By modelling how sun angle and surface albedo interact with topography to produce the observed shading, PFMGAN reduces the confusion between shading caused by slope and shading caused by reflectance variation. Evaluated on five Martian landform classes (impact craters, aeolian dunes, lava flow fields, valley networks, and plains), PFMGAN delivers more than 50% RMSE reduction over baselines across all landform types, with superior performance maintained across spatial scales from 0.25 to 6 m/pixel. The albedo-aware mechanism is directly transferable to the dissertation problem: any generative method that synthesises perspective-view surface appearance from nadir orbital data must similarly disentangle shading caused by geometry from shading caused by intrinsic surface properties. PFMGAN demonstrates this disentanglement is learnable from data with appropriate physical constraints, providing a strong architectural prior.

Cao *et al.* [12] adopt a conditional GAN that maps HiRISE orthorectified images to dense DEMs trained on paired HiRISE orthoimage–DEM data at 2 m and 0.25 m resolution. Evaluation over four geographically distinct Mars regions demonstrates height consistency between predicted DEMs and stereo-derived HiRISE ground truth. The approach is closest in spirit to the dissertation requirement: nadir imagery as input, a 3D representation as output. However, the output is an elevation map, not a perspective-rendered visual texture, and the evaluation geography is limited to four areas with available stereo-based validation data.

Together, Cluster 4 establishes that the geometric component of the problem inferring surface shape from nadir images is tractable with physics-informed deep learning. The component that remains unaddressed is rendering: given a recovered surface geometry, synthesising what that surface would look like from a rover's perspective, with correct photometric properties, realistic regolith texture, and physically plausible shadows.

E. Neural Scene Rendering for Martian Environments

Neural Radiance Fields (NeRF) [17] represent a departure from image-space processing: rather than enhancing or translating 2D images, NeRF methods optimise a continuous volumetric scene function from which any viewpoint can be rendered. This makes NeRF architecturally capable of the view-angle transformation that SR and I2I methods cannot perform.

Giusti *et al.* [13] demonstrate this capability in MaRF, which applies NeRF to multi-image collections from the Curiosity rover at Gale Crater. MaRF confirms that NeRF training converges on Martian imagery despite the unusual photometric environment (persistent dust haze, low solar elevation, high iron oxide spectral contrast), producing novel view synthesis with PSNR and SSIM scores competitive with standard NeRF baselines on Earth scenes. The work establishes Mars rover imagery as a viable NeRF training domain. Its operational limitation is fundamental: MaRF requires multiple overlapping rover images of the *same surface patch* as input. For the ExoMars RFM problem, real images of Oxia Planum do not yet exist, they will be acquired after deployment. Any pre-mission VR training environment must synthesise the rover-perspective view, not reconstruct it from data the rover has not yet taken.

Li *et al.* [14] extend neural Mars rendering into a complete simulation pipeline through the Martian World Model system, comprising two components. The M3arsSynth data engine processes NASA stereo navigation imagery from Curiosity into a multimodal dataset of over 10,000 video clips annotated with depth maps, surface normal estimates, and text descriptions. The MarsGen video diffusion model, trained on this dataset, generates controllable Martian landscape videos conditioned on text prompts, navigation maps, and depth inputs. Experimental results demonstrate that MarsGen surpasses video synthesis models trained on terrestrial datasets in terms of visual fidelity and 3D structural consistency, confirming that domain-specific Mars training data provides significant advantage over transfer from Earth imagery. MarsGen is the most operationally complete system reviewed: it targets mission rehearsal and robotic simulation as explicit use cases, and its controllable generation addresses the requirement that operators can explore different traverse paths in VR. Its limitation is the same as MaRF's, but at larger scale: the entire training corpus is derived from rover perspective imagery. Terrain not previously visited by any rover which encompasses virtually all of Mars outside four landing sites, lies entirely outside MarsGen's generative support.

De Almeida *et al.* [15] address a geometrically adjacent problem: 3D reconstruction from descent-phase wide-angle imagery. The descent trajectory provides a sequence of images spanning nadir to oblique angles, but with strong radial distortion and limited baseline. The paper demonstrates that an implicit neural height field representation outperforms classical multi-view stereo (MVS) under these near-nadir, low-parallax conditions, as the continuous surface prior suppresses depth estimation noise that MVS accumulates. Evaluated on simulated descent sequences over high-fidelity lunar and Mars terrain models, the neural approach achieves better RMSE across a range of terrain roughness levels. This result matters for the dissertation in a specific way: it demonstrates that neural representations with strong surface priors can perform competitively in near-nadir imaging regimes where classical geometric methods fail. If such representations could be coupled to a texture synthesis module, the path from nadir orbital imagery to a rendered perspective scene might run through an intermediate neural height-field representation rather than through explicit 3D mesh construction.

Martínez-Petersen *et al.* [16] integrate NeRF and Gaussian Splatting within a ROS 2-compatible pipeline that processes raw rover rosbag recordings into dense photorealistic 3D reconstructions. The key engineering contribution is Gaussian Splatting's render-time advantage: once trained, a Gaussian scene representation renders at interactive frame rates sufficient for real-time VR, unlike NeRF's ray-marching inference which requires seconds per frame. The system requires perspective-view rover images as its input; no mechanism is proposed or evaluated for deriving the Gaussian scene from orbital data. The finding that Gaussian Splatting achieves VR-compatible render rates is directly relevant to the dissertation's output target, however, as it defines the representation format that the proposed nadir-to-perspective synthesis pipeline should produce.

Paul *et al.* [17] provide a comprehensive survey of NeRF in space applications, identifying terrain mapping, navigation path planning, and mission simulation as the three primary use cases. The survey notes that existing NeRF methods in space contexts uniformly require multi-view imagery as input either rover camera collections or descent image sequences and that the problem of synthesising a NeRF from a single nadir orbital image has not been addressed. This negative finding, reached by the most thorough treatment of the topic, affirms the novelty of the proposed research direction.

F. Virtual Reality Reconstruction and Mission Simulation

Catalano and Soccini [18] address a practical obstacle in Martian VR construction: the heightmaps derived from CTX stereo and MOLA altimetry contain systematic voids arising from instrument shadow, data dropout during transmission, and incomplete orbital coverage. The proposed diffusion model is trained on Mars terrain data and conditioned on the boundary of each void region to inpaint missing elevation values in a structurally consistent manner. Evaluation using SSIM and MS-SSIM against held-out regions of known terrain demonstrates performance gains of up to 15% in RMSE and 81% in perceptual LPIPS score over classical interpolation and prior deep learning baselines. The approach genuinely improves heightmap completeness for VR terrain meshes.

What the method does not produce is surface colour texture. A completed heightmap defines the shape of the terrain mesh but assigns no RGB value to its vertices. A VR environment built from inpainted heightmaps alone renders as a smooth, textureless 3D surface: geometrically faithful but visually sterile and photometrically unrepresentative of what a rover operator would actually see. Texture synthesis, the step that converts a geometric skeleton into a photorealistic surface is acknowledged but not performed. The paper explicitly positions itself as addressing the geometry sub-problem, leaving the texture synthesis challenge open.

Li *et al.* [14] offer the architecturally most complete VR simulation pipeline, but its dependence on rover perspective imagery as its data substrate limits applicability to the four small geographic areas where rover navigation cameras have operated. Pre-deployment simulation for Oxia Planum, a site no rover has visited, lies outside its capability regardless of generation quality.

G. Hallucination and Epistemic Reliability of Generative Models

Every generative model reviewed in the preceding sections can, in principle, produce plausible-looking surface features that have no basis in the underlying physical reality. In natural image applications, invented texture is a quality attribute. In mission-critical planetary science, a synthesised boulder that does not exist or the absence of a real boulder that was not hallucinated which has direct consequences for traversability assessment and crew safety.

Aithal *et al.* [19] provide the most systematic empirical analysis of this failure mode, identifying *mode interpolation* as its proximate mechanism. Diffusion models approximate the score function of the training data distribution with a smooth neural network. When the training distribution contains two distinct modes in close proximity for instance, two crater morphologies that differ in ejecta structure. The smooth approximation assigns non-zero probability to interpolated samples between those modes, which never appeared in training data. Through controlled experiments on synthetic 2D distributions and artificial image datasets, the paper shows that these interpolated samples constitute hallucinations: the model synthesises features that are statistically plausible under its learned score function but have no physical referent in reality. Experiments on real-world image datasets confirm the phenomenon at scale. In the Mars texture synthesis context, this means a diffusion model trained on HiRISE imagery might synthesise impact crater rims with morphologies that do not correspond to any crater at the target site, aeolian ripple wavelengths inconsistent with the local wind regime, or rock textures blending properties of geologically distinct surface units. The scale of this risk is not bounded by model quality alone; it is a structural property of smooth function approximation on multi-modal distributions.

Rathkopf [20] reframes the hallucination problem as an epistemological question: under what conditions can scientists rely on generative model outputs for scientific inference? His answer draws on case studies of AlphaFold 3 (protein structure prediction) and GenCast (atmospheric modelling). In both cases, hallucination risk is converted to bounded, quantifiable risk through two mechanisms: theory-guided training, where domain physical laws impose constraints that prevent the model from generating physically impossible outputs; and confidence-based error screening, where uncertainty estimates flag outputs that should not be used for downstream decisions. Rathkopf argues that generative models embedded in such structured workflows support reliable scientific inference despite their opacity, provided they operate in theoretically mature domains.

The implication for Mars texture synthesis is a concrete design requirement. A generation pipeline that incorporates physical priors Mars solar irradiance models, photometric functions for basaltic regolith, crater morphology distributions derived from impact physics, and aeolian depositional constraints cannot hallucinate a surface feature that violates those priors, because the physics-violation incurs training loss. The dissertation must therefore treat physical constraint not as an optional enhancement but as a primary mechanism for bounding hallucination risk to scientifically acceptable levels.

IV. Quantitative Synthesis and Comparative Analysis

Table I consolidates the twenty reviewed publications across seven comparative dimensions: publication year, methodological approach, task category, input-to-output view transformation, key reported performance metric, whether the method was evaluated on Mars-specific data, and the core limitation that prevents direct application to the nadir-to-rover-perspective VR synthesis problem.

Several patterns emerge from Table I that are not apparent from reading the papers individually. First, a clean partition separates nadir-domain methods (all five SR papers and three DTM papers) from perspective-domain methods (all three NeRF papers that were evaluated on real Mars data). Not a single method crosses this boundary. The view angle of the input determines the view angle of the output in every case except the two theoretical I2I papers, which were benchmarked on natural images only.

Second, the methods with the strongest quantitative evidence of improvement are those that incorporate explicit physical constraints. PFMGAN [11], which embeds solar geometry and albedo modelling into its GAN, achieves more than 50% RMSE reduction over unconstrained baselines, the largest margin of any generation method reviewed. By contrast, the SR papers relying on perceptual loss functions without domain physics (MARSGAN, the early DiffusionSat applications) produce plausible outputs but show greater tendency towards mode-interpolation artefacts of the kind analysed in [19]. This is not a coincidence: physical constraints restrict the generative model's output distribution to a subspace consistent with planetary surface physics, and hallucinated features that lie outside that subspace incur training loss.

Third, the two papers that come closest to a complete VR pipeline MarsGen [14] and the inpainting system [18] each address one dimension of the synthesis problem. MarsGen produces photorealistic perspective-view video but requires rover imagery as input. The inpainting system completes elevation maps from orbital data but produces no surface colour. A system that combines orbital nadir input with perspective-view RGB texture output and is constrained by physical priors does not exist in the reviewed literature.

Fourth, the two I2I papers [9], [10] represent the strongest architectural candidates for closing the nadir-to-perspective

TABLE I
COMPARATIVE OVERVIEW OF REVIEWED LITERATURE AGAINST THE NADIR-TO-ROVER-PERSPECTIVE VR SYNTHESIS PROBLEM. TASK CODES: SR = SUPER-RESOLUTION; I2I = IMAGE-TO-IMAGE TRANSLATION; DTM = DIGITAL TERRAIN MODEL GENERATION; NERF/GS = NEURAL RADIANCE FIELD / GAUSSIAN SPLATTING; VR = VR RECONSTRUCTION PIPELINE; DATA = DATASET CONTRIBUTION; THEORY = THEORETICAL ANALYSIS; REVIEW = SURVEY. VIEW CODES: N = NADIR/ORBITAL; P = PERSPECTIVE/ROVER; 3D = 3D REPRESENTATION; VR_H = VR HEIGHTMAP; "ANY" = DOMAIN-AGNOSTIC.

Ref.	Year	Method	Task	View (I→O)	Key Result	Mars?	Core Limitation for VR Synthesis
[2]	2021	MARSGAN (AW-RRDB)	SR	N→N	≈3× SR on CaSSIS; 8 sites, qualitative	Yes	No quantitative metric; nadir output only; ≈1.3 m/pixel residual gap
[5]	2024	RSTSRN	SR	N→N	×4 SR; +0.88 dB PSNR vs. LapSRN (×4)	Yes	Evaluated on bicubic degradation; real sensor PSF not modelled
[6]	2025	Multi-source Thermal SR	SR	N→N	↑PSNR on THEMIS thermal; multi-modal fusion	Yes	Thermal domain only; no visual RGB texture output
[7]	2024	DiffusionSat	SR/Gen	N→N	SOTA FID on multi-sensor RS; ICLR 2024	No	No Mars training data; Earth geolocation conditioning
[8]	2025	TamingDiff-SRSISR	SR	N→N	×4; ↑PSNR + competitive FID on DOTA	No	Not Mars-specific; no view-angle transformation
[9]	2024	UNSB (Schrödinger Bridge)	I2I	Any→Any	↓FID vs. CycleGAN/CUT; unpaired; ICLR 2024	No	Tested on colour-domain only; no geometry-aware view transform
[10]	2025	Dual-approx Bridge	I2I	Any→Any	Deterministic I2I; ↑SSIM/LPIPS vs. stochastic; CVPR 2025	No	No RS or planetary application demonstrated
[1]	2021	MADNet 2.0	DTM	N→3D	Monocular DTM ≈stereo accuracy; 25 cm/pixel	Yes	Elevation only; no visual surface texture synthesis
[11]	2026	PFMGAN (physics-fusion GAN)	DTM	N→3D	> 50% RMSE↓ over baselines; 5 landform types; 0.25–6 m/pixel	Yes	DTM only; no perspective appearance rendering
[12]	2024	Cond. GAN-DTM (ISPRS)	DTM	N→3D	Height consistency on 4 Mars regions vs. stereo ground truth	Yes	DEM output only; limited geographic validation
[4]	2025	MCTED dataset	Data	N (pairs)	80,898 CTX-DEM pairs; ML-ready	Yes	Nadir+elevation only; no perspective-view samples
[3]	2013	Bayesian rock density (CTX)	Analysis	N→N	Probabilistic rock density from CTX; Bayesian network	Yes	Density inference only; no texture or view transform
[13]	2022	MaRF (NeRF)	NeRF	P→3D	Novel view synthesis; PSNR/SSIM≈standard NeRF	Yes	Requires existing rover images; no synthesis from orbital data
[14]	2025	MarsGen / M3arsSynth	NeRF+VR	P→VR	↑FID vs. terrestrial video models; 10K+ video clips	Yes	Rover perspective data only; unvisited terrain not covered
[15]	2025	Neural height-field (descent)	NeRF	N/oblique→3D	Neural HF>MVS on RMSE (simulated descent)	Partial	Simulated data; descent-phase geometry only; no texture
[16]	2025	NeRF+GS pipeline (ROS 2)	NeRF+GS	P→3D	Real-time GS render; ROS 2 integration; qualitative	Partial	Rover perspective input only; no orbital-to-3D path
[17]	2025	NeRF in space (survey)	Review	—	Comprehensive review; confirms no nadir→perspective method	Yes	Survey only; confirms gap is open
[18]	2025	Diffusion inpainting (VR)	VR	N→VR_H	↑15% RMSE; 81% LPIPS vs. baselines; SSIM/MS-SSIM	Yes	Heightmap only; no RGB texture synthesis
[19]	2024	Mode interpolation analysis	Theory	—	Mechanistic explanation of hallucination; NeurIPS 2024	No	Analytical only; no mitigation method proposed
[20]	2025	Hallucination-to-reliability	Theory	—	Physics-guided training converts hallucination to bounded risk	No	Conceptual framework; no implementation or evaluation

gap, but neither has been evaluated on remote sensing data. The jump from colour-domain translation on natural images to geometry-dependent view-angle transformation on satellite and rover imagery is substantial and constitutes an open research problem.

V. OPEN CHALLENGES AND FUTURE RESEARCH DIRECTIONS

A. The Absence of Aligned Cross-View Training Data

Every supervised learning method reviewed requires either paired training samples (nadir image and corresponding high-resolution target) or at minimum two separate corpora that can be aligned statistically for unpaired training. For the nadir-to-perspective Mars problem, no dataset provides geographically matched nadir and perspective images of the same surface patch at aligned resolution. MCTED [4] provides the largest nadir-

CTX collection to date, but paired with elevation maps rather than perspective imagery. The rover navigation datasets used by MarsGen [14] provide perspective views at four visited sites, with no co-registered nadir counterpart. Assembling a training corpus that bridges both views whether through simulation, multi-view geometric projection, or data augmentation is a methodological prerequisite that none of the reviewed work addresses.

One route is synthetic data: projecting known Mars surface models (derived from HiRISE stereo or from MADNet 2.0 monocular DTMs) into simulated rover perspective renders. This approach is constrained by the realism of the Mars texture models used to colour the geometry and by the domain gap between rendered synthetic scenes and real rover imagery. Narrow-band domain gaps between synthetic and real Mars images have been observed in landing hazard detection contexts [3], and the gap between rendered and real surface photometry is likely wider than domain adaptation methods such as UNSB [9] have been tested on.

B. Hallucination as a Safety-Critical Constraint

Mode interpolation [19] is a structural property of diffusion-based generation, not a defect correctable by training longer or with more data. For Earth-facing applications the consequences of a hallucinated texture are aesthetic. For Mars mission training, a generated crater that does not exist at the landing site, or the absence of a real hazard that the model did not synthesise, could misdirect a crew's traverse planning. The reliability framework of Rathkopf [20] physics-guided training plus confidence-based output screening provides the most credible route to bounded hallucination risk, but it has not been instantiated for planetary imagery generation. Key domain priors that would constrain Mars generation include: cratering density and size-frequency distributions from impact physics; aeolian bedform wavelength–height relationships; photometric functions for basaltic and ferric oxide surface compositions; and shadow angle constraints imposed by Mars's axial tilt and the landing site's latitude and season. Embedding these in the training objective is an open engineering problem.

C. The Computational Cost of Diffusion Synthesis at Planetary Scale

Diffusion models require iterative denoising over many steps (typically 20–1000) per generated sample. DiffusionSat [7] operates on individual tiles at fixed resolution; MarsGen [14] generates video clips of limited spatial extent. Generating continuous, full-site Mars surface texture at rover camera resolution would require either tiling with consistent boundary conditions across tiles, or a cascaded coarse-to-fine approach. Neither strategy has been demonstrated at the scale needed for a mission landing site: Oxia Planum spans roughly 200×200 km, and even a modest 100-pixel-per-metre rover texture resolution implies generating 4×10^{10} RGB pixels. Memory requirements for a diffusion model trained at this scale, and inference times for generating the full canvas, represent unsolved engineering problems. The Gaussian Splatting representation demonstrated in [16] offers one path toward efficient storage and real-time rendering once the scene has been generated, but it does not address the generation bottleneck.

D. Evaluation Benchmarks for Mars Surface Generation

The reviewed SR literature uses at least four distinct evaluation protocols: qualitative geomorphological expert assessment [2], PSNR/SSIM against bicubically downsampled originals [5], perceptual metrics (LPIPS, FID) on remote sensing benchmarks [7], [8], and structural metrics (SSIM, MS-SSIM) on heightmap inpainting [18]. No standardised benchmark exists for evaluating the photorealistic quality and physical correctness of synthesised Mars perspective-view imagery. Without such a benchmark, comparing methods or demonstrating progress towards the VR synthesis objective is not possible in a reproducible manner. Establishing a Mars texture synthesis benchmark with held-out rover images as reference targets and a protocol for generating nadir inputs at matched locations from orbital archives is a necessary community infrastructure contribution.

E. Multi-Modal Fusion Across Instrument Types

The thermal SR work of Lu and Su [6] and the Bayesian rock density prediction of Serrano *et al.* [3] both demonstrate that information from multiple Mars instruments can be combined to recover surface properties unresolvable from any single data source. Visible imaging, thermal IR, and topographic data each carry distinct information about surface composition, thermophysical properties, and physical structure. A generative pipeline that fuses all three using thermal emission to constrain surface material type, topographic derivatives to constrain shape-from-shading ambiguities, and visible reflectance to anchor albedo which would face far fewer hallucination degrees of freedom than a model trained on visible imagery alone. No reviewed paper pursues this multi-modal fusion strategy for texture synthesis, and MCTED's limitation to visible-elevation pairs underscores the data gap this would require addressing.

VI. Conclusion

Twenty publications reviewed here collectively demonstrate that the constituent capabilities needed for VR-grade Mars landing site visualisation, super-resolution of orbital imagery, image domain translation, DTM generation, neural scene rendering, and diffusion-based inpainting have each advanced substantially since 2021. Transformer-based and diffusion-based SR methods now reliably outperform GAN baselines on orbital Mars imagery. Physics-constrained GANs recover Martian terrain geometry with more than 50% lower reconstruction error than their unconstrained predecessors. NeRF and Gaussian Splatting have been demonstrated on real Mars rover data and can, in principle, produce real-time renderable scenes. Diffusion inpainting completes missing elevation maps with quantifiably superior structural fidelity.

What has not been demonstrated is the integration of these capabilities into a pipeline that starts from nadir orbital data and ends with rover-perspective visual texture at VR-grade fidelity. The view-angle transformation from nadir to perspective is the missing operation, and it appears in none of the reviewed methods. The I2I frameworks of [9] and [10] are architecturally the strongest candidates for this transformation, but neither has been applied to remote sensing imagery, let alone to the photometrically complex domain shift between nadir Mars orbital and perspective rover views.

The proposed dissertation targets this specific gap. The analysis of hallucination risk in [19] and the reliability framework of [20] together establish that physics-constrained training is not optional but mandatory for the outputs to be scientifically usable in mission planning. PFMGAN's result [11] that explicit physical priors reduce reconstruction error by more than half which provides a proof-of-principle that such constraints are learnable from data. The combination of unpaired I2I translation (to bridge the view domains without requiring aligned training pairs), physics-informed training objectives (to bound hallucination risk), and a Gaussian Splatting output representation (to enable real-time VR rendering) defines the most promising direction identified by this survey, and it is the direction the proposed work will pursue.

References

[1] Y. Tao, J.-P. Muller, S. Xiong, and S. J. Conway, "MADNet 2.0: Pixel-scale topography retrieval from single-view orbital imagery of mars using deep learning," *Remote Sensing*, vol. 13, no. 21, p. 4220, 2021.

[2] Y. Tao, S. J. Conway, J.-P. Muller, A. R. D. Putri, N. Thomas, and G. Cremonese, "Single image super-resolution restoration of TGO CaSSIS colour images: Demonstration with perseverance rover landing site and mars science targets," *Remote Sensing*, vol. 13, no. 9, p. 1777, 2021.

[3] N. Serrano, P. McGuire, D. Mayer, A. Huertas, and R. Arvidson, "Predicting HiRISE-equivalent rock density on mars using CTX image features," in *AIAA SPACE 2013 Conference and Exposition*. American Institute of Aeronautics and Astronautics, 2013.

[4] R. Osadnik, P. Gómez, E. Bohacek, and R. Bahia, "MCTED: A machine-learning-ready dataset for digital elevation model generation from mars imagery," *arXiv preprint arXiv:2509.08027*, 2025.

[5] F. Wu, X. Jiang, T. Fu, Y. Fu, D. Xu, and C. Zhao, "RSTSRN: Recursive swin transformer super-resolution network for mars images," *Applied Sciences*, vol. 14, no. 20, p. 9286, 2024.

[6] C. Lu and C. Su, "Super resolution reconstruction of mars thermal infrared remote sensing images integrating multi-source data," *Remote Sensing*, vol. 17, no. 13, p. 2115, 2025.

[7] S. Khanna, P. Liu, L. Zhou, C. Meng, R. Rombach, M. Burke, D. B. Lobell, and S. Ermon, "DiffusionSat: A generative foundation model for satellite imagery," in *International Conference on Learning Representations (ICLR)*, 2024.

[8] C. Zhu, Y. Liu, S. Huang, and F. Wang, "Taming a diffusion model to revitalize remote sensing image super-resolution," *Remote Sensing*, vol. 17, no. 8, p. 1348, 2025.

[9] B. Kim, G. Kwon, K. Kim, and J. C. Ye, "Unpaired image-to-image translation via neural Schrödinger bridge," in *International Conference on Learning Representations (ICLR)*, 2024.

[10] B. Xiao, P. Wang, Q. He, and M. Dong, "Deterministic image-to-image translation via denoising Brownian bridge models with dual approximators," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

[11] Z. Zou, Y. Zhou, C. Li, and Z. Dong, "PFMGAN: A generative adversarial network with physics fusion for DTM generation from martian monocular images," *Remote Sensing*, vol. 18, no. 9, p. 1354, 2026.

[12] J. Cao, R. Huang, Z. Ye, Y. Xu, and X. Tong, "Generative 3D reconstruction of martian surfaces using monocular images," in *ISPRS Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. XLVIII-3-2024, 2024, pp. 51–56.

[13] L. Giusti, J. Garcia, S. Cozine, D. Suen, C. Nguyen, and R. Alimo, "MaRF: Representing mars as neural radiance fields," *arXiv preprint arXiv:2212.01672*, 2022.

[14] L. Li, Z. Fan, W. Cong, X. Liu, Y. Yin, M. Foutter, P. Pan, C. You, Y. Wang, Z. Wang, Y. Zhao, M. Pavone, and Y. Wei, "Martian world model: Controllable video synthesis with physically accurate 3D reconstructions," *arXiv preprint arXiv:2507.07978*, 2025.

[15] M. de Almeida, G. Brydon, D. M. Persaud, J. H. Williamson, and P. Henderson, "Neural 3D reconstruction of planetary surfaces from descent-phase wide-angle imagery," *arXiv preprint arXiv:2604.13235*, 2025.

[16] A. Martínez-Petersen, L. Gerdes, D. Rodríguez-Martínez, and C. J. Pérez-del Pulgar, "High-fidelity 3D reconstruction for planetary exploration," *arXiv preprint arXiv:2602.13909*, 2025.

[17] A. Paul, S. A. Deevi, A. Chaudhary, R. K. Lagisetty, Anoop G. L., and Ganesh Kumar R., "Neural radiance fields in space applications: A comprehensive review," *International Journal of Advanced Research*, vol. 13, no. 3, pp. 525–559, 2025.

[18] G. L. Catalano and A. M. Soccini, "Inpainting the red planet: Diffusion models for the reconstruction of martian environments in virtual reality," *arXiv preprint arXiv:2510.14765*, 2025.

[19] S. K. Aithal, P. Maini, Z. C. Lipton, and J. Z. Kolter, "Understanding hallucinations in diffusion models through mode interpolation," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 37, 2024.

[20] C. Rathkopf, "From hallucination to reliability: Generative modeling and the structure of scientific inference," *arXiv preprint arXiv:2504.08526*, 2025.